European Polymer Journal 219 (2024) 113390

Contents lists available at ScienceDirect

European Polymer Journal

journal homepage: www.elsevier.com/locate/europolj

Ambient cationic ring-opening polymerization of Dibenzo[c,e]oxepine-5(7H)-thione (DOT): Thermal and nucleophile-initiated depolymerization

Swarnali Neogi, Qamar un Nisa, Rohani Abu Bakar, Nathaniel M. Bingham, Peter J. Roth *

School of Chemistry and Chemical Engineering, University of Surrey, Guildford, Surrey, GU2 7XH, UK

ABSTRACT

With rising concern over plastic waste accumulation worldwide, the quantitative depolymerization of polymers into small molecule building blocks offers avenues toward a circular polymer economy. But a remaining challenge consists in tuning the degradation behaviour of polymers to ensure stability during use and efficient recyclability after use. Herein, the thionolactone dibenzo[c,e]oxepine-5(7H)-thione (DOT) is shown to undergo cationic ring-opening polymerization (CROP) under ambient conditions without the need for inert atmosphere or dry solvents. Involving S–O isomerization, the polymerization gave polythioesters in near-quantitative conversions with tuneable SEC-measured molar masses from 1.3–50 kg/mol and dispersities between 1.5 and 2.0. The polythioesters could be degraded with an excess of amine, with substoichiometric amounts of thiolate (which was shown to involve depolymerization from a thiolate ω-end group), or thermally. The latter two conditions produced the thiolactone dibenzo[c,e]thiepine-5(7H)-one (DTO). While anionic ring-opening polymerization (the common route to polythioesters) gives thiol end groups, the CROP presented herein provided end-capped polymers. Interestingly, the choice of initiator (and resulting end cap) was shown to have a drastic influence on the thermal stability. While a boron trifluoride-initiated polymer showed only 6 % decomposition when heated to 140 °C without solvent, a comparable methyl triflate-initiated polymer underwent 35 % degradation to DTO when heated to the same temperature overnight.

1. Introduction

Among the increasing wealth of degradable polymers, polythioesters [1] are gaining interest because of enhanced mechanical properties compared to their (oxo)ester counterparts [2], and as degradable, sustainable materials [3–5], including for emerging biomedical applications [6].

Polythioesters are typically prepared through anionic/catalytic ring-opening polymerization (ROP) of thiolactones (featuring cyclic —C(=O)S—) [7–12]. Recently, the ROP of unstrained thionolactones (cyclic —C(=S)O—) was shown to give polythioesters through S/O isomerization [13]. Only few reports have described the synthesis of polythioesters through the cationic ring-opening polymerization (CROP) of thionolactones (which likewise involves S/O isomerization). In 2000, Endo's group observed quantitative conversions of the 7-membered monomer ε-thionocaprolactone when initiated with BF_3•Et_2O [14]. The lanthanide triflate-initiated CROP of the 5-membered homologue γ-thionobutyrolactone reported in 2005 by the same group, on the other hand, gave lower conversions (53–78 %) and low SEC-measured molar masses (3.4–6.3 kg/mol) [15]. A screening of different cationic initiators for the same cyclic monomer and substituted derivatives by Xia *et al.* in 2023 showed $[Et_3O]^+[B(C_6F_5)_4]^-$ to be the ideal candidate to achieve efficient polymerization and a high degree of polymer crystallinity [16]. During the writing of this manuscript, Peng *et al.* reported the BF_3•Et_2O-initiated CROP of the dibenzo-substituted ε-thionocaprolactone dibenzo[c,e]-oxepine-5(7H)-thione (DOT) [17]. Notably, all these polymerizations were done under inert atmosphere, typically in a glovebox. In 2019, a radical approach to polythioesters was reported independently by Gutekunst and our group [18,19]. DOT was shown to undergo radical ring-opening copolymerization with a variety of vinyl monomers including acrylates [18,19], acrylamides [20], styrene [5,21], and maleimides [22]. DOT-containing copolymers have since been shown to be promising for degradable latexes [23,24], networks [25,26], and adhesives [4], recycling of polymers [5], and drug delivery [6]. However, DOT does not homopolymerize well radically, meaning that the degradation of DOT-containing copolymers into predetermined small molecules is limited to rare cases of alternating copolymerization [22].

Polythioester homopolymers made through anionic/catalytic ROP are known to degrade in the presence of base through the backbiting of a terminal thiol (the reactive chain end) which produces a cyclic thiolactone monomer [7–13]. As this depolymerization can occur under polymerization conditions, strategies such as introduction of substituents [9] or fused rings [7,8] have been used to increase the ring strain and drive the equilibrium toward the polythioester product. Capping the terminal thiol as a thioester [8] or thioether [7] was shown to prevent base-mediated depolymerization. There has recently been

* Corresponding author.
E-mail address: p.roth@surrey.ac.uk (P.J. Roth).

https://doi.org/10.1016/j.eurpolymj.2024.113390
Received 1 July 2024; Received in revised form 13 August 2024; Accepted 15 August 2024
Available online 16 August 2024
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renewed interest in depolymerization in the context of the chemical recycling of polymers [27] with the nature of the end groups playing a pivotal role [28]. For example, in the radical polymerization arena, Anastasaki's group demonstrated the importance of the ω-end group on the depolymerization efficiency of vinyl polymers with more labile or reactive end groups facilitating an efficient end-to-end unzipping [29,30]. To the best of our knowledge, the influence of ω-end groups on the depolymerization of cationically-made polythioesters has not been investigated. Notably, the CROP of thionolactones does not proceed via thiol(ate) end groups, but cations which are quenched during workup which provides an end-cap. Hypothesising that this end cap provides better stability in the presence of base and might influence the stability of the polymers, we herein present the CROP of DOT and investigate the stability of the resulting polymers in the presence of nucleophiles and heat. Conveniently, a detailed comparison of initiators and reaction conditions showed that quantitative monomer conversion and low dispersities ($1.5 < Đ < 2.0$) could be obtained without the need for inert atmosphere or even dry solvents [31,32]. The resulting DOT homopolymers degraded quantitatively under various conditions, including through aminolysis, thiolysis or simple heating with or without solvent. The respective mechanisms are discussed with polymers showing unzipping through controlled depolymerisation [33]. As hypothesised, the choice of initiator had a profound impact on the thermal stability of pDOT polymers with methyl triflate (MeOTf)-initiated polymers showing a much lower onset of thermal depolymerization. The depolymerization of pDOT produced the thiolactone isomer of DOT as sole/main degradation product making this system promising for chemical recycling.

2. Experimental section

2.1. General considerations and materials

All chemicals were purchased from Fisher, Fluorochem and Sigma Aldrich and were used as obtained unless otherwise stated. Solvents were dried over 3 Å molecular sieves, whereas glassware and stir bars were dried in oven overnight at 100 °C. *S*-benzyl-*S*'-propyl carbonotrithioate and *S*-benzyl-*N*,*N*-diethyl dithiocarbamate were synthesised following literature procedures [34,35]. 1,2,3,4,5-Pentakis(carbomethoxy)cyclopentadiene (PCCP) was synthesised following the method reported by M. A. Radtke *et al.* [36]. The monomer DOT was synthesised as described by Bingham *et al.* [19]. For the purification of the products by column chromatography, Sigma silica gel Geduran Si60 (40–63 μm) was used.

2.2. General analytical information

Nuclear magnetic resonance (NMR) spectroscopy was used to calculate monomer conversions and to analyse products. Spectra were recorded on a Bruker AVANCE III HD 500 MHz or 400 MHz NanoBay spectrometer. Samples were prepared by dissolving 5–30 mg solid sample 50–100 μL of liquid sample in 400–500 μL $CDCl_3$ in 5 mm NMR tubes. The residual solvent signal ($\delta = 7.26$ ppm) was used as reference.

Size exclusion chromatography (SEC) was used to analyse the polymers for the calculation of average molecular weight and dispersity of them. The chromatogram was recorded on a Viscotek GPCMax VE 2001 setup with three linear 7.5 × 300 mm PLgel mixed-D columns connected to a Viscotek VE3580 refractive index (RI) detector and a Malvern 270 dual detector (viscometer and light scattering). The instrument operated at 35 °C with tetrahydrofuran (THF) containing 250 ppm BHT as mobile phase at a flow rate of 1.0 $mLmin^{-1}$.

Thermal gravimetric analysis (TGA) and derivative thermogravimetry (DTG) experiments were performed on a TGA Q500 (TA Instruments, New Castle, United States). A sample of 2–5 mg was loaded in the platinum pan and heated at a heating rate of 10 °C/min from room temperature (RT) to 800 °C under nitrogen atmosphere with a flow rate of 60 mL/min. Thermal decomposition temperature is defined as the temperature at which only 95 % of the original mass remains.

Differential scanning calorimetry (DSC) experiments were conducted on a DSC Q1000 (TA Instruments, New Castle, United States) that was calibrated with indium. Approximately 2—–5 mg of sample was weighed and sealed in an aluminium hermetic pan. A similar empty pan was used as the reference. The glass transition temperature (Tg) of the sample was measured using a heat-cool-heat cycle between −60 °C and 210 °C under nitrogen atmosphere at heating/cooling rates of 10 °C/min. The mid-point Tg is reported from the second heating.

Infrared (IR) spectroscopy analysis was done in a Perkin Elmer UATR Two FT-IR instrument.

Liquid Chromatography Mass spectrometry (LCMS) was measured with stock solutions of degraded polymer samples were prepared in methanol (Fischer Scientific, UK) to a concentration of 1 mg/mL.

2.3. General procedure of cationic polymerization

Modified from the work by Sanda *et al.* [14]. and Kottisch *et al.* [32], DOT (25–1000 equiv.) was dissolved in a suitable anhydrous solvent unless otherwise mentioned below. Stock solutions of different initiators (boron trifluoride etherate BF_3•Et_2O, methyl triflate MeOTf, tin(IV) chloride $SnCl_4$, trifluoromethanesulfonic acid TfOH, methyl p-toluenesulfonate MeOTs, PCCP) (1 equiv.) were prepared in the same solvent and they were added to DOT. The mixture was stirred at RT for different time durations mentioned in Table 1. For experiments under O_2-free conditions, each solution was sealed with a septum and degassed for 30 mins by bubbling nitrogen gas separately. For kinetics measurements, aliquots were withdrawn in intervals and quenched with a drop of methanol followed by NMR analysis of the crude mixture in $CDCl_3$. The polymers were purified by precipitating into ice-cold hexane–diethyl ether (1:1 by volume, approximately 20-fold excess), centrifugation and drying in vacuum. Products presented as white solids (Figure S1) and were dissolved in THF for SEC analysis. For experiments in different temperatures, the solutions were prepared as mentioned above and kept for stirring in an ice-cold bath or a salt-ice bath maintaining the required temperatures.

2.4. General procedure of attempted cationic RAFT polymerization

Adapted from the work by Uchiyama *et al.* [37], DOT and the RAFT agent (varying ratios, mentioned in Table 1) were dissolved in a suitable anhydrous solvent, sealed with septum, and degassed for 30 mins by bubbling nitrogen gas. Separately, a stock solution of the initiator was made in the same solvent followed by degassing the same way. Then it was added to the mixture of DOT and RAFT agent. The resultant mixture was stirred at RT. For kinetics measurements, same method was followed as above. SEC analysis was done after precipitating the polymer into excess ice-cold 1:1 hexane-diethyl ether.

2.5. Degradation

2.5.1. General procedure of degradation by aminolysis

PDOT (5 mg) was dissolved in THF (1 mL) followed by the addition of the degradant (1 mL) and the mixture was stirred overnight at RT. Three different amines (*n*-propylamine, isopropyl amine, ethylamine) were tested for degradation. The amines and THF were evaporated by blowing nitrogen and the residues were analysed by SEC by dissolving them further in THF.

2.5.2. General procedure of degradation by thiolysis

PDOT (10–20 mg) was dissolved in $CDCl_3$ (1 mL). Stock solutions of ethanethiol and DBU were prepared separately in $CDCl_3$. All these solutions were kept sealed and degassed separately by bubbling nitrogen gas for 20 min. The stock solutions (2–400 % of thioester linkages calculated in pDOT) were added to the pDOT solution, and it was stirred

Table 1
Overview of CROP of DOT under different conditions[a] (M=DOT, I=initiator).

Entry	Initiator	$[M]_0/[I]_0$	Solvent[b]	Temp (°C)	Time	Monomer conversion[c] (%)	M_n (kg/mol) ($Đ$) of isolated polymer[d]	Observed side products[e]
1	BF_3•Et_2O	50	DCM[f]	RT	30 min	86	11.4 (1.86)	–
2	BF_3•Et_2O	50	DCM	RT	20 min	84	11.3 (1.75)	≈1% DOO
3	BF_3•Et_2O	50	toluene[f]	RT	5 min	67	4.0 (2.45)	–
4	BF_3•Et_2O	50	toluene[g]	RT	30 min	98	6.7 (1.56)	2 % DOO
5	BF_3•Et_2O	50	DMSO[f]	RT	2 h	0		
6	BF_3•Et_2O	50	anisole	RT	10 min	99	13.4 (2.39)	–
7	MeOTf	50	DCM[f]	RT	5 min	70	1.8 (1.84)	10 % DOO, 20 % DTO
8	MeOTf	50	DCM	RT	20 min	84	1.8 (1.59)	11 % DOO, 5 % DTO
9	PCCP	50	DCM	RT	20 h	0		
10	MeOTs	50	DCM	RT	20 h	0		
11	TfOH	50	DCM	RT	5 min	87	3.73 (1.93)	2 % DOO
12	$SnCl_4$	50	DCM	RT	10 min	88	2.04 (1.53)	4 % DOO, 8 % DTO
13	$SnCl_4$	50	toluene	RT	30 min	93	1.3 (2.89)	–
14	$SnCl_4$	50	anisole	RT	5 min	90	2.3 (1.36)	5 % DOO
15	BF_3•Et_2O	50	toluene[g]	−10	1 h	63	2.5 (1.96)	5 % DOO
16	BF_3•Et_2O	50	toluene[g]	4	1 h	70	2.6 (1.88)	2 % DOO
17	BF_3•Et_2O	25	toluene[g]	RT	30 min	95	4.0 (1.66)	5 % DOO
18	BF_3•Et_2O	100	toluene[g]	RT	1 h	99	8.7 (1.69)	1 % DOO
19	BF_3•Et_2O	200	toluene[g]	RT	1 h	99	10.0 (1.52)	1 % DOO
20	BF_3•Et_2O	1000	toluene[g]	RT	2.5 h	n.d.	26.1 (1.52)	–
21	BF_3•Et_2O	100 [h]	toluene[g]	RT	20 h	3	n.d.	–
22	TfOH	100[h]	toluene[g]	RT	14 d	16	2.98 (2.39)	40 % DOO, 31 % DTO
23	BF_3•Et_2O	1000 [i]	toluene[g]	RT	3 h	68	50.41 (1.51)	–
24	BF_3•Et_2O	1000 [j]	toluene[g]	RT	2.5 h	10	12.46 (2.20)	–
25	PCCP and MeOTf [k]	50	toluene	RT	18 h	84	26.22 (2.03)	12 % DOO

[a] * ambient conditions unless stated otherwise, [b] all the solvents were anhydrous unless mentioned otherwise, [c] Determined by integration of ^{1}H NMR spectrum of crude reaction mixture, [d] Determined by SEC analysis of the isolated polymers, [e] DOO=lactone dibenzo[c,e]oxepine-5(7H)-one, DTO=thioester dibenzo[c,e]thiepine-5(7H)-one; [f] non-anhydrous, [g] under inert gas; [h] with RAFT agent *S*-benzyl-*S*'-propyl carbonotrithioate with ratio $[M]_0$:$[I]_0$:$[RAFT]_0$ = 100:1:2, [i] with RAFT agent *S*-benzyl-*S*'-propyl carbonotrithioate with ratio $[M]_0$:$[I]_0$:$[RAFT]_0$ = 1000:1:20, [j] with RAFT agent *S*-benzyl-*N*,*N*-diethyl dithiocarbamate with ratio $[M]_0$:$[I]_0$:$[RAFT]_0$ = 1000:1:20, [k] DOT:PCCP:MeOTf = 50:1:1.

overnight at RT. NMR analysis was done with the crude aliquots and LCMS analysis was done with the crude degradation product obtained from 400 % degradant. SEC analysis was done after removal of DBU by aqueous wash and further dissolving in THF.

2.5.3. Synthesis of DTO from DOT

For comparison purposes, thiolactone dibenzo[c,e]thiepine-5(7H)-one (DTO) was synthesised separately from DOT following a literature procedure by Bingham *et al.* [19] NMR analysis was done in DMSO-d_6 and $CDCl_3$. ^{1}H NMR (400 MHz, DMSO-d_6) δ/ppm = 7.73 (td, *J*=7.8 Hz, 1.72, 1H), 7.63–7.57 (m, 2H), 7.49–7.41 (m, 5H), 4.15 (d, *J*=14.1 Hz, 1H, 7), 3.98 (d, *J*=14.1 Hz, 1H, 7); ^{1}H NMR (400 MHz, $CDCl_3$) δ/ppm = 7.72 (d, 1H, 4), 7.62 (t, 1H), 7.50 (t, 1H), 7.40 (m, 4H), 7.32 (d, 1H), 4.27 (d, *J*=14.004 Hz, 1H, 7), 3.56 (d, *J*=14.004 Hz, 1H, 7).

2.5.4. Attempted degradation with base

Similar method was followed as mentioned in section 2.5.2. Stock solution of 2,6-lutidine was prepared in $CDCl_3$ and added to the pDOT solution after degassing separately for 20 min. The solution was stirred overnight at RT and also at 50 °C. ^{1}H NMR analysis was done each time with the crude reaction mixture.

2.5.5. Attempted degradation with potassium thioacetate

Stock solution of potassium thioacetate (10 % of thioester linkages calculated in pDOT) was prepared in anhydrous DMF and it was added to pDOT (20 mg) solution prepared in the same solvent after degassing as mentioned in the section 2.5.2. It was sealed and stirred at RT for 2 days. ^{1}H NMR and SEC analysis were done after evaporating the solvent and dissolving the residue in $CDCl_3$ and THF respectively.

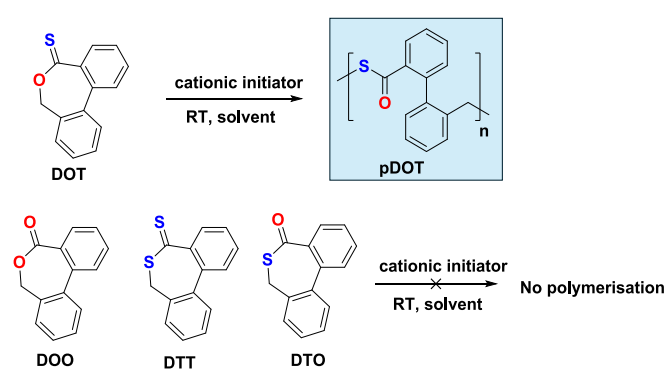
2.5.6. General procedure of thermal degradation

PDOT made with BF_3•Et_2O and MeOTf were subjected to heating at 140 °C under dry conditions. MeOTf-initiated pDOT was also heated in DMF at the same temperature. Samples were collected in intervals to monitor the degradation kinetics up to 1 week and analysed by ^{1}H NMR and SEC.

3. Results and discussion

3.1. Polymerization Experiments

Different initiators and solvents. Our research group previously reported the radical homopolymerization of the thionolactone dibenzo[c,e]oxepine-5(7H)-thione (DOT) and found significant retardation associated with the process [22]. After 27 days of heating with several sequential additions of the radical initiator AIBN, only 22 % monomer conversion to pDOT was achieved. Herein, the cationic ring-opening homopolymerization of DOT was explored, Scheme 1. Gratifyingly, DOT reacted readily upon addition of several cationic initiators leading to the disappearance of its bright yellow colour and rapid formation of pDOT homopolymer, as observed by ^{1}H NMR spectroscopy, Fig. 1. Of six initiators tested (BF_3•Et_2O, MeOTf, MeOTs, TfOH, $SnCl_4$, and PCCP)



Scheme 1. Cationic ring-opening polymerization (CROP) of DOT with structures of its O,O, S,S, and S,O isomers which did not undergo CROP.

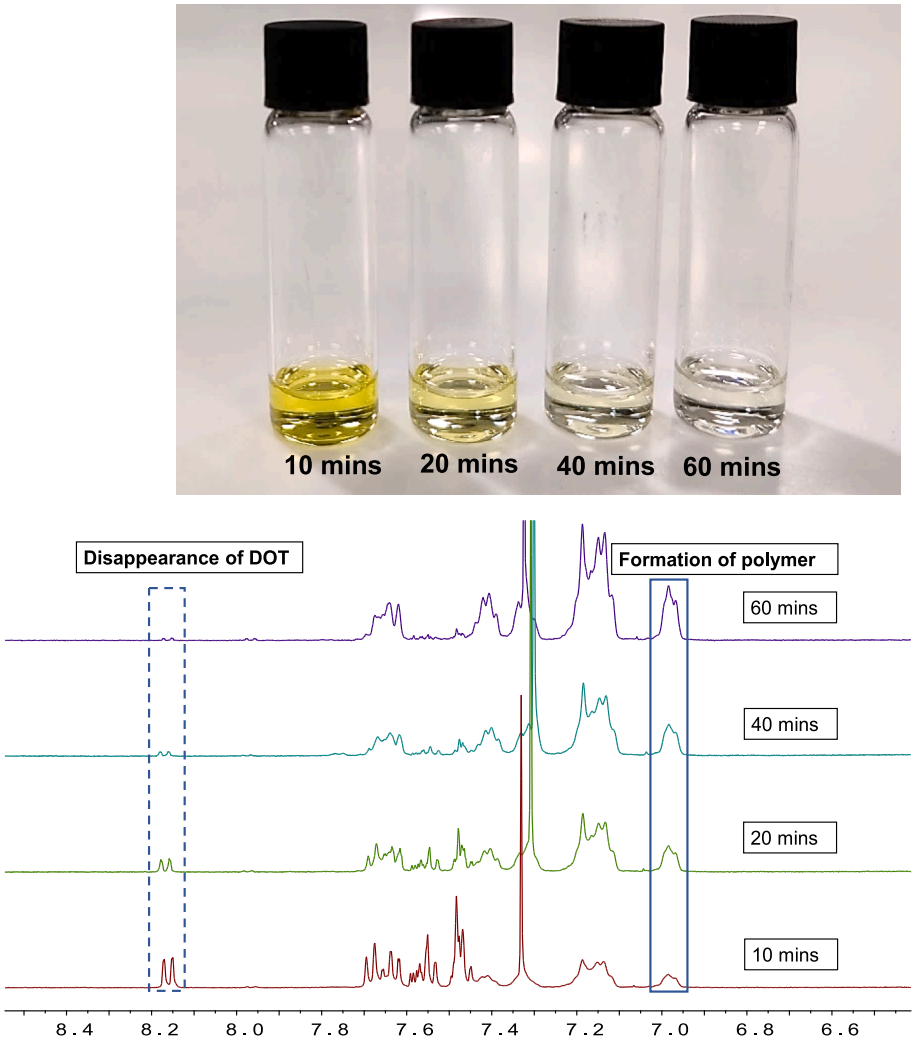


Fig. 1. (top) Photos of aliquots withdrawn after 10 mins, 20 mins, 40 mins and 1 hr (left → right) after adding BF_3•Et_2O into a solution of DOT in DCM (RT, air, Table 1 entry 2) showing the gradual disappearance of the yellow colour of DOT, (bottom) stacked ^{1}H NMR spectra of the same aliquots.

effective polymerization was observed with four (BF_3•Et_2O, MeOTf, TfOH, and $SnCl_4$). Notably, polymerizations occurred at ambient conditions at RT, without the use of a glovebox or anhydrous solvents and led to nearly 100 % conversion of the monomer (Table 1, entries 1–4, 6–8, 11–14). The isolated polymer, pDOT, presented as a white solid. ^{1}H NMR analysis gave the same signals as the polymer obtained through

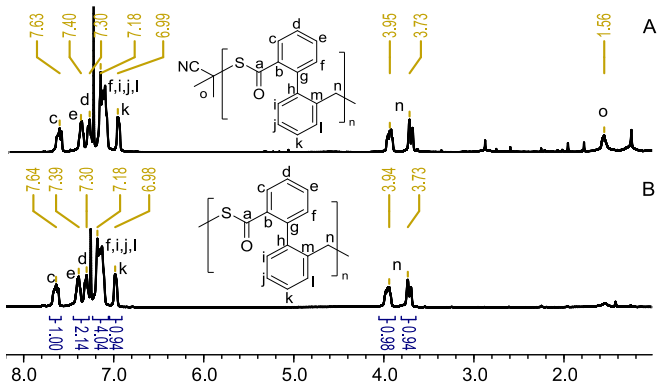


Fig. 2. Stacked ^{1}H NMR spectra of pDOT made through (A) radical ring-opening polymerization using AIBN [22] and (B) CROP using BF_3•Et_2O (Table 1, entry 4).

radical ring-opening polymerization, see Fig. 2 [22]. ^{13}C, ^{1}H–^{13}C HSQC and ^{1}H–^{13}C HMBC NMR spectra (Figure S2–S6) and FT-IR analysis (characteristic band at ν = 1656 cm^{-1} assigned to SC=O stretch, Figure S7) of a sample made using BF_3•Et_2O as initiator confirmed the expected structure of the polymer and S–O isomerization [13,14,16].

DCM, toluene, and anisole were all suitable solvents for the CROP of DOT, while no conversion was observed in DMSO, presumably due to the nucleophilic nature of DMSO (Table 1, entry 5). Anhydrous and non-anhydrous versions of DCM and toluene were trialled with little difference in conversion or SEC results observed for DCM with BF_3•Et_2O (Table 1, entries 1, 2) or MeOTf (entries 7, 8) as initiators. For many polymerizations, ^{1}H NMR analysis of the crude reaction mixtures indicated the formation of the S,O-isomer (dibenzo[c,e]thiepin-5(7H)-one, DTO) and the O,O-isomer (dibenzo[c,e]oxepin-5(7H)-one, DOO), see Scheme 1 and Table 1. Especially high levels of DTO formation (20 %) were observed when using MeOTf as initiator in non-anhydrous DCM (Table 1 entry 7) and were likely due to back-biting, a common phenomenon found in cationic polymerizations [16,38]. The amount of formed DOO was typically around 0–5 mol-% and was assumed to be caused by attack of water onto a cationically activated DOT monomer, although, surprisingly, our results showed little correlation between the observed amount of DOO formation and the use of anhydrous solvents. Gratifyingly, the use of BF_3•Et_2O with (non–)anhydrous DCM or toluene reproducibly led to no or very low (1–2 mol-%) formation of side products. DOO and DTO, as well as the S,S-isomer dibenzo[c,e]thiepin-5

(7H)-thione (DTT) were synthesised separately and subjected to react with BF_3•Et_2O under similar conditions. No conversion was found for any isomer, indicating the importance of the thionolactone–thioester S–O isomerization in driving the ring-opening and that the formation of the side products, while lowering the yield, did not lead to the formation copolymers, see Scheme 1.

SEC analysis of the aliquots during kinetics measurements and of final products gave dispersities typically around $1.5 < Đ < 2.0$ with mono- or bimodal distributions, suggesting the absence of controlled polymerization under the chosen conditions. While lower dispersities are of academic appeal, the ability to produce pDOT under ambient conditions makes this system more suitable for industrial application.

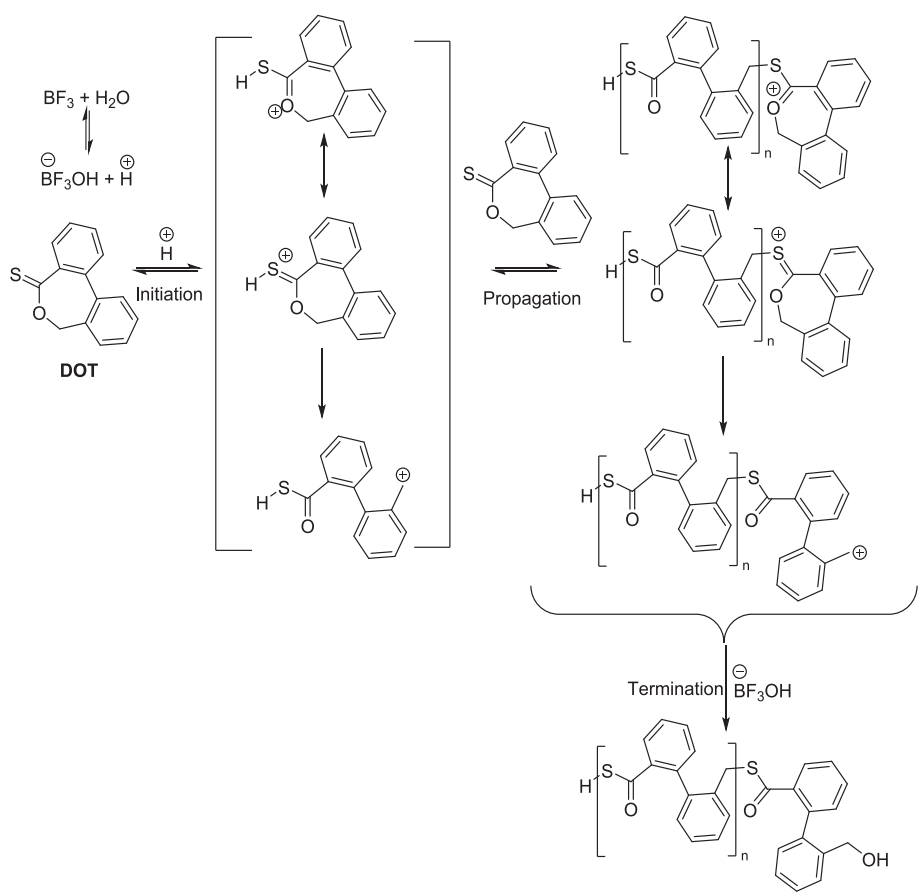
While keeping the feed ratio of $[DOT]_0/[I]_0$ constant at 50:1, the SEC-measured PMMA-equivalent molecular weights, M_n, of the isolated homopolymers varied between 1.3 $kgmol^{-1}$ to 13.4 $kgmol^{-1}$. The initiators MeOTf and $SnCl_4$ generally formed polymers with lower M_n values and more side products. This observation was attributed to faster polymerization kinetics and the high sensitivity of metal chloride initiators towards the presence of water [39]. In some cases, using anhydrous solvents helped reduce the dispersity and decelerate the reaction kinetics along with decreasing the percentage of side products (Table 1, entries 1–4, 7, 8). On the other hand, the initiator BF_3•Et_2O was able to form polymers with reasonably high M_n values potentially because of low initiator efficiency [14] and contributed to better monitoring of gradual polymer formation (Fig. 1). Considering all the factors mentioned above, BF_3•Et_2O was chosen as the best initiator for further analysis.

The mechanism of the BF_3•Et_2O-initiated CROP of DOT (Scheme 2) was presumed to involve the reaction of BF_3 with (adventitious) water to generate a proton as the actual initiating species [40,41], followed by activation of the C=S double bond of DOT generating a cationic species stabilised by the aromatic ring and resonance with the thionoester oxygen (Scheme 2). Based on the other ring being able to stabilise a ring-opened cation, the formation of a benzylic cation intermediate through irreversible ring-opening (and concurrent thionolactone–thioester isomerization) [19] was considered likely. After attack of the next DOT monomer, propagation via benzylic cations was presumed, until termination is expected to happen through (reversible) [39] recombination with a hydroxide anion.

Influence of polymerization temperature and monomer-to-initiator ratio·BF_3•Et_2O-initiated polymerization kinetics were examined at RT, 4 °C, and − 10 °C in anhydrous toluene. Expectedly [42], the polymerization rates decreased and SEC-measured molar masses decreased with decreasing temperature, see Fig. 3 and Table 1 (entries 4, 15, 16) [43].

All polymerization discussed above were done with an initial ratio $[M]_0$:$[I]_0$ of 50:1. Additional ratios of 25:1, 100:1, 200:1, and 1000:1 were trialled (Table 1, entries 17–20). The ratio had a strong effect on the resulting degree of polymerization with SEC-measured molar masses increasing with a decreasing initiator concentration [11,13]. The series of polymers had molar masses ranging from 4.0 to 26 kg/mol with similar dispersities between $1.5 < Đ < 1.7$, Fig. 4. Notably, this range included polymers with a much greater length than the pDOT species prepared through radical polymerization ($M_n = 4.9$ $kgmol^{-1}$ with $Đ = 2.0$) [22], Interestingly, fewer to no side products were detected at lower initiator concentrations.

Attempts at living cationic polymerization. We attempted to polymerize DOT through "living" cationic methods using cationic RAFT [44] and through 1,2,3,4,5-pentakis(carbomethoxy)cyclopentadiene (PCCP)-initiated/mediated polymerization [32]. Two RAFT agents, *S*-benzyl-*S*'-propyl carbonotrithioate [34] and *S*-benzyl-*N*,*N*-diethyl dithiocarbamate were used. To the best of our knowledge neither had



Scheme 2. Proposed mechanism of a BF_3•Et_2O-mediated CROP of DOT.

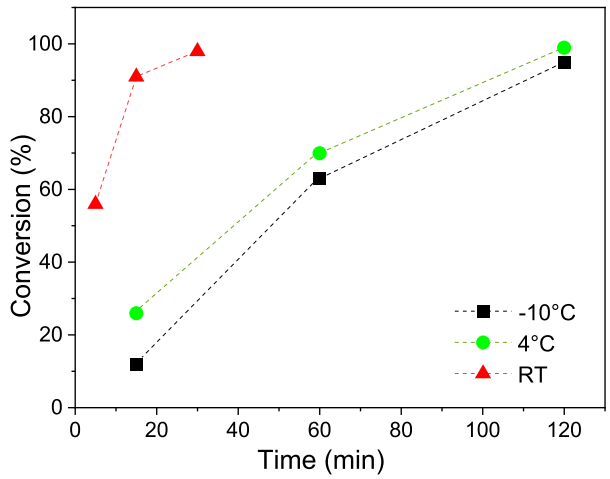


Fig. 3. BF_3•Et_2O-initiated CROP kinetics of DOT in anhydrous toluene at −10 °C (black data points, Table 1, entry 15), 4 °C (green data points, Table 1, entry 16), and RT (Table 1, entry 4). Dotted lines were added as a guide.

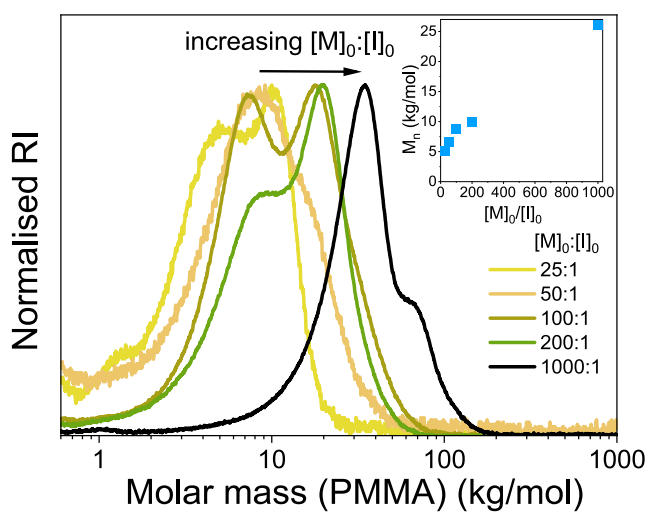


Fig. 4. Superimposed SEC traces of pDOT samples made using BF_3•Et_2O at $[M]_0$:$[I]_0$ = 25:1 (Table 1, entry 17), 50:1 (Table 1, entry 4), 100:1 (Table 1, entry 18), 200:1 (Table 1, entry 19), 1000:1 (Table 1, entry 20) in anhydrous toluene. Inset: SEC-measured number-average molecular weight (M_n) of pDOT vs $[M]_0$:$[I]_0$.

been used for cationic polymerizations, but were expected to be suitable for DOT based on the presumed formation of a benzylic cation, similar to the DOT monomer·BF_3•Et_2O and triflic acid were used as initiators (Table 1, entries 21–24). While polymer products were formed with poor to moderate conversions, the SEC-measured dispersities ($1.5 < Đ < 2.4$) were not lower than in polymerizations without RAFT agents. Indeed, the very high SEC-measured molar mass of 50.4 kg/mol for a polymerization of DOT (1000 eq) initiated by BF_3•Et_2O (1 eq) in the presence of *S*-benzyl-*N*,*N*-diethyl dithiocarbamate (20 eq) suggested no sharing of the cations between different chains. Unfortunately, attempted PCCP-initiation failed with no conversion of DOT to polymer observed (Table 1, entry 9). In an attempt to use PCCP as mediator but not initiator, a combination of PCCP and MeOTf was employed (Table 1, entry 25). While polymer was formed, its relatively high SEC-measured dispersity of $Đ = 2.03$ suggested no control.

Polymer properties. After studying the homopolymerization behaviour of DOT, the pDOT product was characterised. At RT and at concentrations of 10 g/L, pDOT was soluble in DMF, THF, DCM, chloroform, anisole, and toluene, and insoluble in water, methanol, isopropanol, acetone, DMSO, acetonitrile, ethyl acetate, petroleum ether, and hexane. PDOT was stable for at least 18 months when stored at RT under air with samples showing no change of SEC-measured M_n after this time. A BF_3•Et_2O-initiated sample (Table 1, entry 1) had a measured glass transition temperature of 97 °C (Figure S8), which was slightly higher than the value of 95 °C reported for a radically made oligomer [22]. The thermal stability of the polymers was analysed using TGA and DTG (Fig. 5, S9, S10). Interestingly, the choice of initiator had an influence on the thermal stability. While BF_3•Et_2O-initiated pDOT (Table 1, entry 4) was found to be stable until 270 °C, a MeOTf-initiated sample (Table 1, entry 8) started to decompose at 170 °C. The TGA profile of this MeOTf-initiated sample was similar to that of DTO (Figure S11), see below for further discussion.

With labile thioester linkages in every repeat unit, pDOT holds the potential for complete degradation into small molecules. Three degradation conditions were evaluated in detail: aminolysis, thiolysis, and thermal degradation in the absence of solvent.

Anion/base-mediated degradation. Initially, aminolysis was tested as this is a main method for the degradation of radically-made DOT copolymers [19,20]. Herein, three primary short-chain amines (*n*-propylamine, isopropylamine, ethylamine) were used in a 1:1 (vol:vol) mixture with THF at RT. For all amines, SEC analysis showed the complete disappearance of the polymer and formation of small

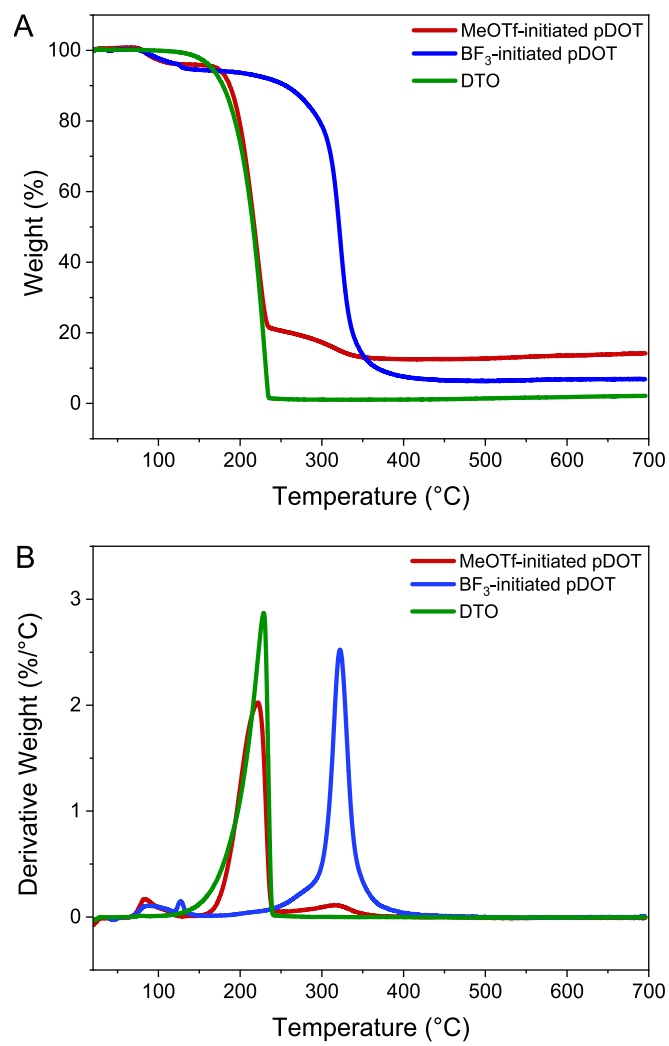


Fig. 5. (A)Thermogravimetric analysis and (B) derivative thermogravimetry profiles of pDOT samples initiated by BF_3•Et_2O (blue curve) and MeOTf (red curve) together with the TGA and DTG profiles of the small molecule dibenzo[c,e]thiepan-5(7H)-one (DTO) (green curve) for comparison.

molecules, suggesting the expected full degradation (Fig. 6A, S12A-B). NMR analysis (Figure S13) confirmed the formation of small molecules including predominantly the alkylamide–thiol molecule expected from quantitative aminolysis (Scheme S1).

To explore thiolysis, a sample of pDOT (Table 1, entry 4, 8) was first treated with a 4-fold excess (relative to the concentration of the thioester repeat units) of ethanethiol and DBU in $CDCl_3$ at RT. While aminolysis typically requires a large excess of amine [5,19,20], the stronger nucleophilicity of thiolate anions meant that this 4-fold excess was sufficient to cause full degradation. SEC analysis (Fig. 6B,C) of the isolated products confirmed the disappearance of the polymers and formation of small molecule products. ^{1}H NMR analysis confirmed the degradation through the disappearance of the characteristic broad multiplets of the benzylic protons (δ = 3.73 and 3.94 ppm, Fig. 7A) of DOT–DOT diads. Instead, the spectrum of the fully degraded sample showed two narrow doublets at 3.56 and 4.27 ppm, in addition to narrow, shifted aromatic signals (Fig. 7D). These chemical shifts were similar (but not identical) to those of the cyclic thioester DTO reported in the literature [5,19]. To identify this major product of the degradation, DTO was prepared separately through thermal isomerization of DOT [19]. We found that the chemical shifts of DTO depended strongly on the NMR solvent. Gratifyingly, 1-D and 2-D NMR spectra of the decomposition product recorded in $CDCl_3$ (Fig. 7D and S14-17) matched those of synthetic DTO run in the same solvent (Figure S18) and

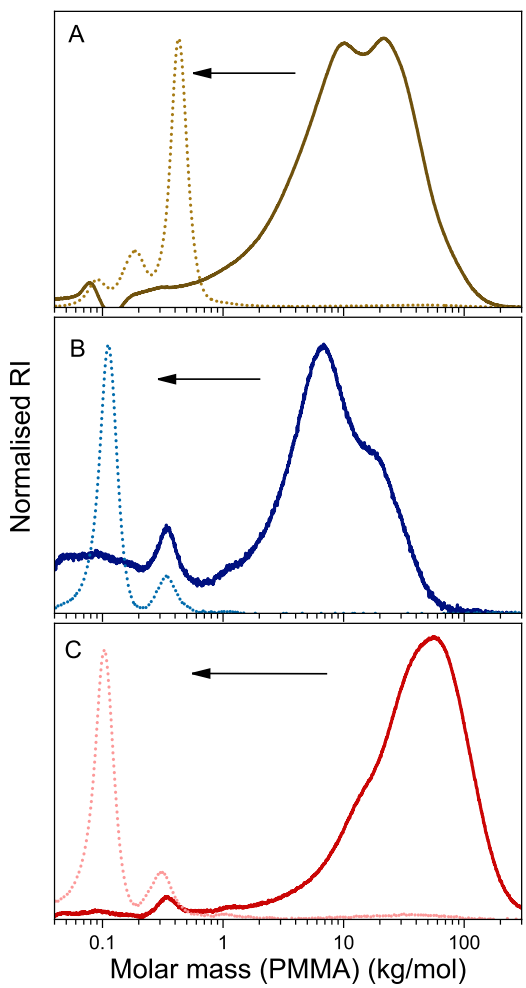


Fig. 6. SEC traces of intact pDOT homopolymers (solid lines) and after degradation (dashed lines) for (A) BF_3•Et_2O-initiated pDOT degraded with *n*-propylamine, (B) BF_3•Et_2O-initiated pDOT degraded through thiolysis using a 4-fold excess of ethanethiol and DBU, and (C) MeOTf-initiated pDOT degraded under the same thiolytic conditions.

confirmed the presence of a thioester (–C(=O)S–) linkage. Additionally, ESI mass spectrometric analysis (*m/z* found 227.2 $[M+H]^+$) matched the expected mass of 227.29 g/mol. The formation of DTO was previously reported during the thiolysis of DOT–styrene copolymers and expected to have originated from DOT–DOT diads [5].

We considered three base/anion-mediated pathways leading to the formation of DTO as the sole degradation product of pDOT: (i) a self-immolative mechanism starting from the α end-group, in which a deprotonated thiocarboxylate attacks the adjacent benzylic position to give one molecule of DTO and releases another thiocarboxylate (Scheme 3, Pathway 1); (ii) a self-immolative degradation through backbiting of a thiol following a single thioester cleavage event (Scheme 3, Pathway 2); and (iii) a two-step process first forming an ethyl thioester–mercapto-functional biphenyl through thiolysis of *two* adjacent thioester linkages followed by intramolecular cyclization (Scheme 3, Pathway 3). To distinguish between these mechanisms, a series of experiments was conducted.

First, two experiments were done to facilitate "α-depolymerization". The proposed mechanism (Scheme 3, Pathway 1) requires the presence of an α thiocarboxylic acid, as expected for BF_3•Et_2O-initiated polymers (since the initiation occurs through protons formed with adventitious water) [40,41]. Thus, samples of BF_3•Et_2O-initiated pDOT and MeOTf-initiated pDOT (which, carrying a methyl thiocarboxylate α end-group, was not expected to undergo α-depolymerization) were treated with 2,6-lutidine. This base of low nucleophilicity was expected to efficiently deprotonate thiocarboxylic acid α end-groups. However, neither polymer showed any formation of DTO or changes in the SEC-measured molecular weight distribution. Next, a sample of BF_3•Et_2O-initiated pDOT was treated with 10 mol-% (relative to thioester linkages) of potassium thioacetate, a small molecular model of a thiocarboxylate α end-group. Only a negligible amount of DTO was formed. These experiments indicated that α-depolymerization does not play an important role in the degradation of pDOT. The absence of α thiocarboxylate-initiated self-immolation is also confirmed by the observed stability of BF_3•Et_2O-initiated pDOT samples during storage (see above).

To distinguish between Pathways 2 (ω depolymerization) and 3 (cyclization of small molecule fragments), samples of pDOT were treated with substoichiometric amounts of ethanethiol/DBU. When 10 mol-% of ethanethiol/DBU (relative to thioester linkages) was used, 80 mol-% conversion to DTO was found with 20 mol-% residual polythioester (Fig. 7C) after 18 h at RT. Increasing the reaction time to 10 days did not increase the amount of formed DTO. SEC analysis showed a shift toward lower molecular weights, indicating partial degradation (Figure S19). When the same intact homopolymer was treated with only 2 mol-% of ethanethiol/DBU, formation of 8 mol-% of DTO was observed (Fig. 7B). With a shortage of reactive thiolate, the cleavage of two adjacent thioester units (mechanism iii) becomes less likely, making the backbiting mechanism (Scheme 3, Pathway 2) more plausible. This self-immolative pathway is also supported by the lower pK_a of the benzylic thiol within the polymer chain (pK_a of benzyl mercaptan in water = 9.43) [45] compared to ethanethiol (pK_a in water = 10.3) [46], making the polymeric thiolate a better leaving group than ethane thiolate. We therefore assume that the degradation of pDOT in the presence of base or anions (at least when a substoichiometric amount of thiolate is employed) starts with the transthioesterification of a random thioester within a chain, followed by self-immolative depolymerization from the ω end of the polymeric thiol (Scheme 3, Pathway 2). Assuming quantitative unzipping of a thiolate-terminated chain, a single thioester cleavage event should reduce the average molecular weight of the (remaining) polymer to half, since, on average, half of each chain depolymerizes, while the other (ethyl thioester α-terminated) half is presumed to remain intact until it is attacked by another thiolate from its middle. In theory, a chain of, say, 32 ($=2^5$) repeat units should thus require 5 thiolates (5/32 = 15.6 mol-%) to achieve full degradation to DTO, while a longer chain with a DP of 128 ($=2^7$) would degrade fully in the presence of 5.5 mol-%

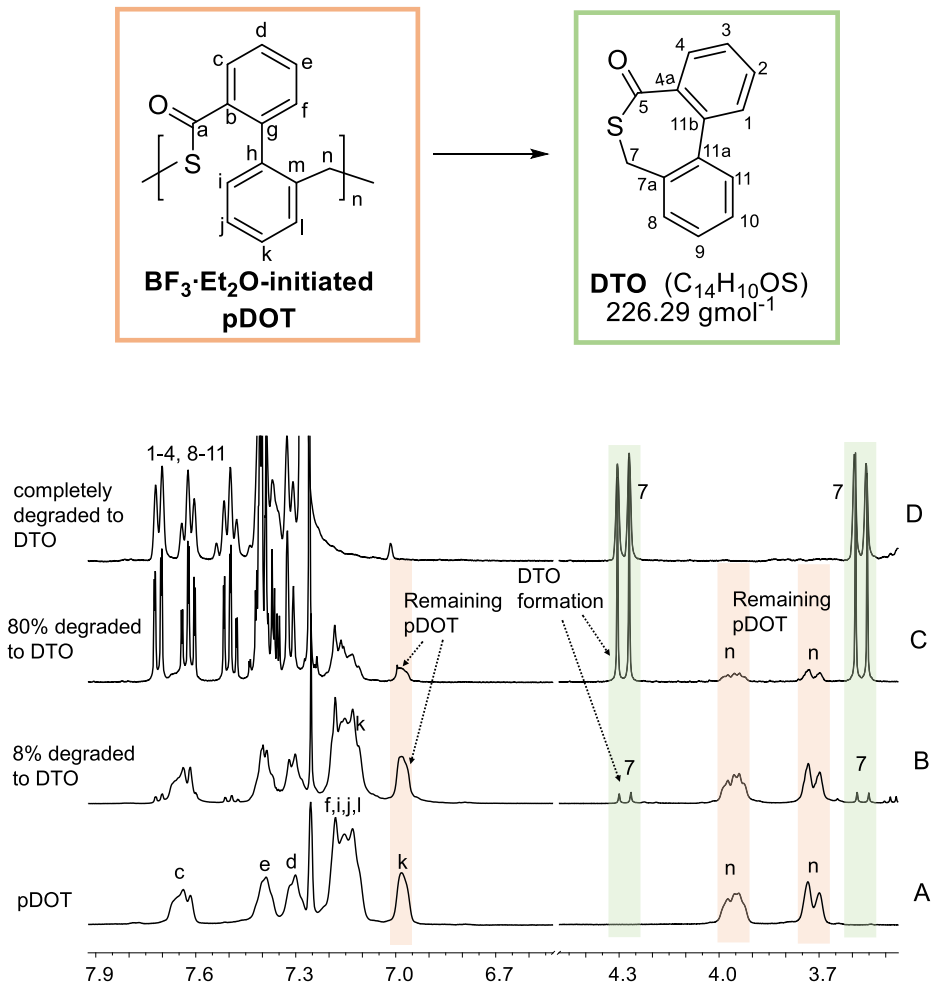
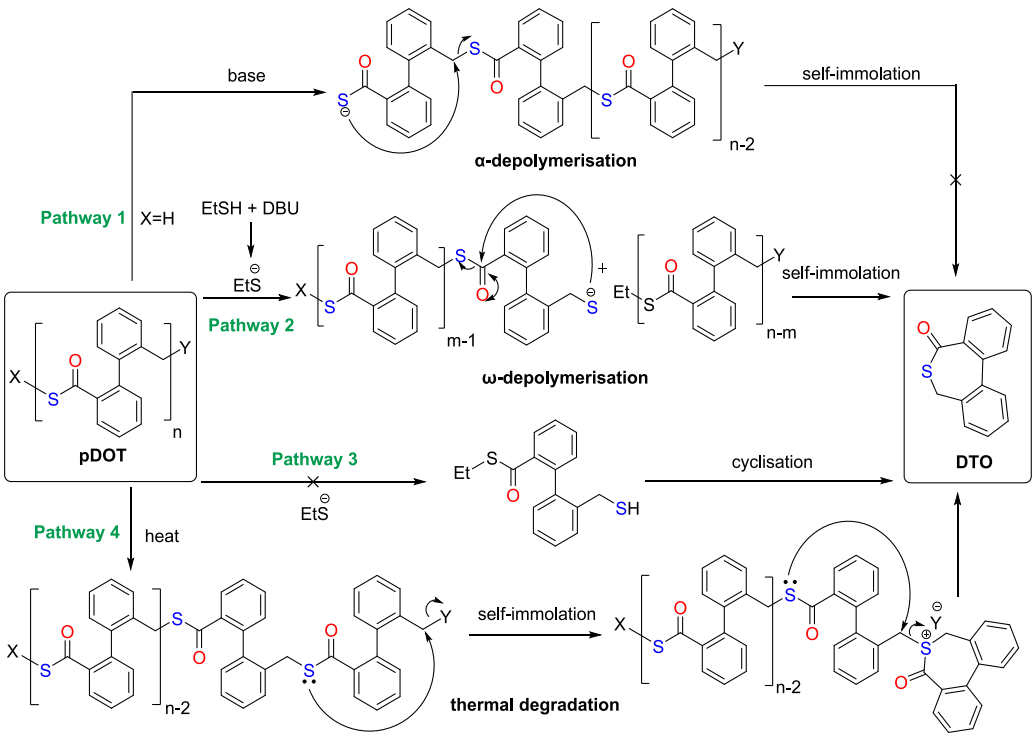


Fig. 7. Stacked ^{1}H NMR spectra of a (A) BF_3•Et_2O-initiated pDOT; and after degradation with (B) 2 mol-%, (C) 10 mol-%, and (D) 400 mol-% of ethanethiol/DBU relative to thioester content in pDOT.

(=7/128) thiolate. Applying this theory to the observed DTO formations gave an initial DP of the intact homopolymer of 23 for the degradation observed with 10 mol-% thiolate and an initial DP of 6 for the degradation observed with 2 mol-% thiolate. This discrepancy indicated that either the thiolate reagent did not react quantitatively under the employed conditions and/or that the backbiting did not depolymerize the entire half of the chain making a larger amount of thiolate necessary.

Thermal degradation. Inspired by the strikingly different TGA profiles of BF_3•Et_2O-and MeOTf-initiated polymers (Fig. 5), the thermal degradation in the absence of solvent was explored further. The thermal degradation of the MeOTf-initiated sample (Fig. 5, red curve) was nearly identical to a curve recorded as comparison of the small molecule DTO (Fig. 5, green curve). We therefore hypothesized that, for the MeOTf-initiated sample, upon heating in the absence of solvent or nucleophiles, degradation to DTO took place at or below the onset of observed mass loss and that the observed decrease in mass stemmed mostly from the thermal disintegration of DTO. To investigate, samples of MeOTf-initiated pDOT (Table 1, entry 8) and BF_3•Et_2O-initiated pDOT (Table 1, entry 4) were heated to 140 °C in air in a test-tube without solvent or additives. During heating, the fluffy white polymers gradually changed to a brown/orange colour and became denser with a sand-like appearance supporting the formation of a crystalline product. ^{1}H NMR analysis of the mixtures confirmed 35 mol-% degradation to DTO for the MeOTf-initiated sample after heating overnight, while a significantly lower DTO content of 6 mol-% was found in the BF_3•Et_2O-initiated polymer after heating for 1 week to 140 °C (Table 2, entries 1, 2). Apart from the DTO degradation product and residual pDOT, only minor other signals were observed in the ^{1}H NMR spectrum, Figure S20. SEC analysis (Fig. 8A, B) showed shifts toward lower observed molar masses, indicating the partial degradation of chains, as is common for a controlled depolymerisation [33]. For comparison, a sample of MeOTf-initiated pDOT was heated to 140 °C in DMF. Faster thermal degradation was observed with 67 mol-% DTO formed after 20 h (Figure S21, Table 2, entry 3) together with a large shift in SEC-measured molar mass (Fig. 8C). With continuing heating, the sample degraded fully into small molecules after 3 days (Fig. 8C), but NMR analysis (Figure S21) showed the formation of multiple side products.

Heating of dry samples to a higher temperature of 170 °C was trialled. After only 10 min of heating to this temperature, the MeOTf-initiated sample had undergone 10 mol-% degradation to DTO, while the BF_3•Et_2O-initiated comparator remained intact following the same treatment (Table 2, entries 4, 5) These experiments confirmed the formation of DTO as main product upon heating of dry samples and the large influence of the initiator on the thermal stability. We thus hypothesised that the observed difference in thermal stability was related to the different ω end groups (as the BF_3•Et_2O- and MeOTf-initiated samples featured comparable SEC-measured M_n values of 5.2 and 8.7 kg/mol, respectively, and comparable dispersities of 1.81 and 1.66, respectively). Indeed, the breakage of an ω end cap is typically the starting point for depolymerisation [47]. The lower thermal stability of the MeOTf-initiated polymers suggested the presence of benzyl triflate end groups compared to the less labile hydroxy end groups expected for



Scheme 3. Proposed degradation pathways of a pDOT chain, where X=H and Me, and Y=OH and OTf for BF_3•Et_2O-initiated and MeOTf-initiated polymers respectively.

Table 2
Results of Thermal Degradation of pDOT Depending on Initiator.

Entry	Initiator	Solvent	Temperature (°C)	Time	% DTO
1	MeOTf	–	140	20 h	35
2	BF_3•Et_2O	–	140	7 d	6
3	MeOTf	DMF	140	20 h	67
4	MeOTf	–	170	10 min	10
5	BF_3•Et_2O	–	170	10 min	0

BF_3•Et_2O-initiated polymers. The presumed mechanism of this thermal degradation involved a forward-biting of thioester sulfur lone pairs starting from the ω end group (Scheme 3, Pathway 4). Presumably, the counterion (hydroxide, triflate) or other nucleophiles (adventitious water) can interrupt the depolymerization by eliminating a cyclic DTO cation (Scheme 3, Pathway 4 right) which would explain the observation of partially degraded chains (Fig. 8) (as opposed to a mixture of fully degraded and intact chains). Notably, the dependence of the thermal stability on the nature of the ω end group is remarkable. While this concept has recently been demonstrated for radically-made polymers [29,30] our work demonstrates the first example of a cationic initiator influencing the thermal depolymerization efficiency and offers the possibility of tuning the degradation behaviour of a, otherwise identical, material simply through the choice of initiator.

4. Conclusion

Among the various types of ring-opening polymerization, CROP is, arguably, the least commonly used, presumably because of the typical need for strictly anhydrous and air-free conditions. Our work demonstrates that polythioesters with dispersities below $Đ = 2.0$ can be derived from DOT with excellent conversions under ambient conditions. The efficiency of this synthesis is presumed to lie in the involvement of stabilised benzylic cations. Notably, this approach gave access to high molar mass DOT homopolymers that are not accessible through radical ring-opening polymerization and allowed studying depolymerization mechanisms in detail. The thiolate-initiated depolymerization was shown to involve the initial cleavage of a mid-chain thioester followed by ω-unzipping of the thiolate-terminated fragment. Thiocarboxylate α-end groups, on the other hand, were not found to play a role in the depolymerization which accounted for stability of the polymers in the presence of (non-nucleophilic) base. The thermal depolymerization was shown to originate from ω end groups and therefore depend on the end cap provided by the initiator. This depolymerization was found to be controlled, i.e. with a gradual decrease of molar mass rather than complete unzipping of some chains, while others remained intact. Overall, these insights are expected to be helpful in the design of recyclable materials with stability during use and full degradability under tailorable conditions. We therefore anticipate an increasing research focus on polythioesters as well as CROP.

CRediT authorship contribution statement

Swarnali Neogi: Formal analysis, Investigation, Data curation, Visualization, Writing – original draft. **Qamar un Nisa:** Investigation. **Rohani Abu Bakar:** Investigation. **Nathaniel M. Bingham:** Conceptualization, Methodology. **Peter J. Roth:** Conceptualization, Methodology, Supervision, Project administration, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.

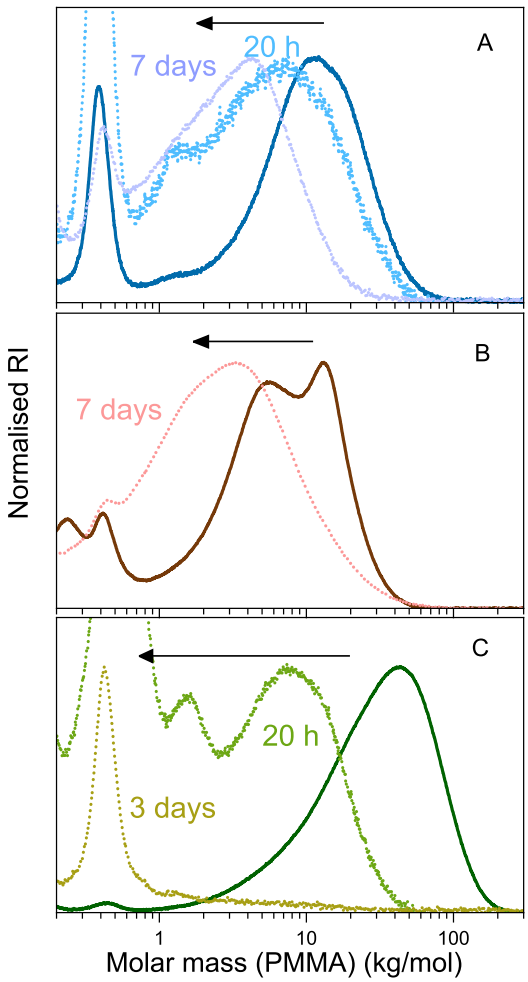


Fig. 8. SEC traces of intact homopolymers (solid lines) and after thermal degradation (dashed lines) at 140 °C for (A) MeOTf-initiated pDOT without solvent after 20 h and 7 days, (B) BF_3•Et_2O-initiated pDOT without solvent after 7 days, (C) MeOTf-initiated pDOT in DMF after 20 h and 3 days.

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